You're absolutely right, JR — this is the moment to step back and treat your project not just as a coding exercise, but as a real product design challenge. What your lecturer told you is gold: you need to understand the business mechanics of recruitment before you can build a system that works in practice.

Here's how you can structure your research so it's actionable:

Research Areas

1. How companies currently recruit WIL students

- **Large companies: Often use HR departments, structured graduate programs, or partnerships with universities.**
- **SMMEs (small/medium enterprises): Usually rely on word-of-mouth, local networks, or ad-hoc postings. They rarely have formal recruitment pipelines.**
- **Institutions: Many universities already have career offices or WIL coordinators who act as the middleman.**

2. Pain points companies face

- **Admin overload: Paperwork, contracts, compliance with labor laws.**
- **Matching skills: Hard to filter students by relevant qualifications.**
- **Trust & vetting: Companies worry about whether students are prepared or reliable.**
- **Compliance: In South Africa, POPIA and labor regulations add complexity.**

3. Incentives for companies

- **Visibility: Access to a pool of vetted, qualified students.**
- **Reduced effort: Automated matching, pre-verified documents, less admin.**
- **Reporting: Easy dashboards showing placements, compliance, and outcomes.**
- **Partnership benefits: Recognition by institutions, possible tax incentives or government support.**

Stakeholder Map

- **Students: Need placements to graduate. They want clarity, deadlines, and verified opportunities.**

- **Companies: Need talent but don’t want extra admin. They want efficiency and trust.**
- **Institutions: Need reporting, compliance, and improved graduation rates. They want control and oversight.**

Refined Value Proposition

Instead of:

“A place where students upload docs and companies post programs.”

Frame it as:

“A platform that helps institutions streamline WIL placements, gives companies easy access to vetted candidates, and ensures students graduate on time.”

This way, you’re solving a shared pain point for all three parties, not just students.

Here’s a clear written outline of the hybrid workflow that balances institutions, companies, and students without over-centralizing control:

🔗 Hybrid WIL Placement Workflow

Step 1 – Student onboarding

- **Students upload academic records, CVs, and WIL eligibility documents into the platform.**
- **Institution verifies these documents (academic status, compliance, readiness).**
- **Verified students are marked as “eligible” in the system.**

Step 2 – Company requirements

- **Companies submit WIL program requirements (number of students, skills, duration, location).**
- **They don’t directly browse raw student profiles — instead, they see a pool of vetted candidates.**
- **Companies retain the right to shortlist and interview.**

Step 3 – Institution as validator

- **Institution ensures only qualified students are matched to company requirements.**
- **They act as a compliance gatekeeper (making sure placements meet academic and legal standards).**
- **They generate reports for graduation tracking and accreditation.**

Step 4 – Matching process

- **Platform auto-matches students to company requirements based on verified data.**
- **Institution reviews and approves the match.**
- **Companies receive a shortlist of vetted candidates.**

Step 5 – Company selection

- **Companies conduct interviews or assessments.**
- **Final selection is made by the company, not the institution.**
- **Institution records the placement outcome in the system.**

Step 6 – Reporting & outcomes

- **Institution tracks placements, graduation timelines, and compliance.**
- **Students see their placement status clearly.**
- **Companies get dashboards showing their WIL involvement and student progress.**

⚖️ Why this works

- **Students: One trusted hub, less risk, clear graduation pathway.**
- **Institutions: Control over verification and compliance, but not overloaded with recruitment admin.**
- **Companies: Access to vetted candidates, but still in charge of final hiring decisions.**

This hybrid model avoids the bottleneck of institutions acting as full recruiters, while still solving the fragmentation and trust issues you identified.

Here's a database schema workflow outline for the hybrid WIL placement model we discussed. It's structured to reflect the three stakeholders (students, institutions, companies) and the matching process:

📄 Core Entities

Students

- student_id (PK, UUID)
- name, surname
- email, phone
- academic_record (link/file reference)
- program_of_study
- graduation_year
- verified_status (boolean, set by institution)

Institutions

- institution_id (PK, UUID)
- name
- contact_person
- email
- reporting_requirements (JSON or text)

Companies

- company_id (PK, UUID)
- name
- industry
- contact_person
- email
- verified_status (boolean, set by institution)

📄 WIL Programs / Requirements

WIL_Program

- program_id (PK, UUID)
- company_id (FK → Company)
- title
- description
- required_skills (array/text)
- slots_open (int)
- duration_months
- open_date, close_date
- status (enum: open, closed, filled)

📝 Matching & Applications

Applications

- application_id (PK, UUID)
- student_id (FK → Student)
- program_id (FK → WIL_Program)
- status (enum: pending, shortlisted, selected, rejected)
- date_applied

Institution_Match

- match_id (PK, UUID)
- institution_id (FK → Institution)
- program_id (FK → WIL_Program)
- student_id (FK → Student)
- match_status (enum: proposed, approved, rejected)
- date_matched

📝 Placements & Reporting

Placements

- placement_id (PK, UUID)
- student_id (FK → Student)
- program_id (FK → WIL_Program)
- company_id (FK → Company)
- institution_id (FK → Institution)
- start_date, end_date
- completion_status (enum: ongoing, completed, failed)

Reports

- report_id (PK, UUID)
- institution_id (FK → Institution)
- program_id (FK → WIL_Program)
- placement_id (FK → Placement)
- graduation_impact (boolean)
- notes

🔄 Workflow Logic

1. Student registers → Institution verifies eligibility.
2. Company posts WIL program requirements.
3. Institution matches students → creates Institution_Match records.
4. Company reviews shortlist → updates Application.status.
5. Final selection → recorded in Placements.
6. Institution generates reports → tracks graduation outcomes.

This schema ensures:

- Students are vetted before applying.
- Institutions act as validators and matchmakers.
- Companies retain final hiring control.
- Placements and Reports provide accountability and graduation tracking.

👉 **Do you want me to draw up an ERD in PlantUML syntax for this schema so you can visualize the relationships clearly and share it with your team?**

You're right to scrutinize this, JR — the flow of verification is just as important as the schema. Let's break down how it would actually work in practice, given that institutions already hold student records:

🔄 **Student Application & Verification Flow**

Step 1 – Student onboarding into the system

- **Students don't "apply" with raw documents — instead, the system pulls their existing records (academic transcripts, enrollment status, WIL eligibility letters) directly from the institution's database.**
- **The student confirms or updates personal details (CV, contact info, skills).**

Step 2 – Institution verification

- **Since the institution already has the records, verification is essentially a flagging process:**
  - **"Eligible for WIL" → student has completed required coursework and submitted a WIL letter.**
  - **"Not eligible" → student hasn't met prerequisites.**
- **This avoids duplicate uploads and ensures authenticity.**

Step 3 – Company posts requirements

- **Company submits a WIL program (slots, skills, duration).**
- **Institution reviews the posting for compliance (e.g., stipend rules, duration requirements).**

Step 4 – Matching

- **System auto-matches eligible students to company requirements.**
- **Institution validates the shortlist (ensuring only qualified students are forwarded).**

Step 5 – Company selection

- **Companies interview or assess shortlisted students.**
- **Final decision rests with the company.**

Step 6 – Placement recording

- **Institution records the placement outcome in the system.**
- **This feeds into graduation tracking and reporting.**

⚖️ Why this matters

- **Students don't waste time uploading documents the institution already has.**
- **Institutions act as validators, not recruiters — they confirm eligibility and compliance.**
- **Companies still control final hiring, but benefit from vetted candidates.**

So the truth is: students don't "apply" in the traditional sense. They're pre-verified by the institution, then matched to opportunities. The institution's role is to validate and track, not to manually re-collect data they already own.

👉 Do you want me to outline how this flow would look in terms of API endpoints (e.g., /students/verify, /programs/post, /matches/create) so you can start thinking about how to implement it in Supabase/Express?

Good question — let's be practical about what a final-year student actually does before they register on a WIL portal like TUT's.

🎓 Student Flow Before WIL Registration

1. Institution already has records

- **The university holds your enrollment data, academic transcripts, and progress toward graduation.**
- **You don't need to re-upload those — they're already in the system.**

2. Student provides "placement-ready" info

- **What the institution doesn't have (or doesn't maintain in a recruitment-friendly format) are:**

    - CV/resumé
    - Portfolio links (GitHub, LinkedIn, etc.)
    - Skills profile (specific technologies, soft skills)
    - Availability (dates you can start WIL)
- These are what you, as the student, would enter into the WIL system.

3. Institution verifies eligibility

- The system cross-checks your student ID against the university database.
- If you've completed the required coursework and submitted a WIL letter, you're flagged as "eligible."
- This verification is automatic — no manual uploads of transcripts needed.

4. Student applies to programs

- Once verified, you can express interest in WIL programs posted by companies.
- Your application is backed by institutional verification, so companies trust the data.

5. Institution tracks placement

- When you're matched or selected, the institution records it in the system.
- This ensures your placement counts toward graduation requirements.

⚖️ Key Distinction

- Institution's role: Verify eligibility, ensure compliance, track graduation.
- Student's role: Provide CV, skills, and preferences — the "human" side of employability.
- Company's role: Post requirements, interview, and select candidates.

So to answer directly: as a final-year student, you don't dump all your academic records into the WIL system — the institution already has those. You only add the career-oriented information (CV, skills, availability) that the institution doesn't store in detail.

👉 Do you want me to outline a student dashboard design (fields, sections, actions) that shows exactly what a student would see and fill in when they log into such a WIL portal? That would make this flow more concrete.

You've spotted a real structural problem in how recruitment cycles and academic calendars clash. Companies often recruit early in the semester (March/April or September/October), while students only get their WIL letters/transcripts after exams — meaning many miss the window. That's not just inconvenient; it's systemic misalignment.

⚠️ The Catch

- **Companies' timeline: They want interns/WIL students lined up before the semester ends, so they can onboard in January or July.**
- **Students' timeline: They only become "officially eligible" after exams, results, and WIL letters are issued.**
- **Result: Students who finish late in the semester miss the recruitment cycle and have to wait 6+ months.**

🛠️ How to Cover for This Gap

1. Conditional Eligibility

- **Institutions can issue a "provisional WIL letter" once a student is registered for final modules.**
- **It states: *"Pending successful completion of coursework, this student is eligible for WIL placement."***
- **Companies can recruit early, knowing placements are conditional on passing.**

2. Rolling Verification

- **Instead of waiting for all results, institutions can verify students module by module.**
- **If a student has completed 90% of requirements, they're flagged as "eligible pending final exam."**

3. Placement Reservation

- **Companies can reserve slots for students who are "provisionally eligible."**

- Final confirmation happens once transcripts/WIL letters are issued.
- This prevents students from missing the cycle entirely.

4. Institution–Company Agreements

- Institutions can negotiate with companies to accept conditional candidates earlier.
- This aligns recruitment with academic calendars, instead of forcing students to wait until January.

5. System Design Implication

- Your platform should support two statuses:
  - Provisional eligibility (student is finishing coursework, awaiting final results).
  - Verified eligibility (student has transcript + WIL letter).
- Companies see both, but know provisional students are subject to final clearance.

🎯 The Truth

If you don't build this two-tier eligibility system, your platform will replicate the same bottleneck students already face — missing recruitment cycles. Institutions must provide conditional verification earlier, otherwise the system won't solve the timing mismatch.

👉 Do you want me to outline how to model these two statuses in your database schema (e.g., eligibility_status = provisional | verified) so you can implement it cleanly in Supabase?

...

You're thinking exactly like a software engineer should — looking for every loophole that could break the system for WIL students. Let's list the major ones beyond the timing mismatch we already discussed, so you can design around them:

🔍 Loopholes & Risks in WIL Placements

1. Recruitment Timing vs Academic Calendar

- **Companies recruit early, students only get WIL letters after exams.**
- **Fix: Introduce *provisional eligibility* status so students can be considered before final transcripts are issued.**

2. Verification Bottleneck

- **Institutions may delay issuing WIL letters or transcripts due to admin backlogs.**
- **Fix: Automate verification by linking the WIL system directly to the institution's academic database. Reduce manual paperwork.**

3. Company Incentives

- **SMMEs may not see value in hosting WIL students (extra admin, no clear benefit).**
- **Fix: Provide reporting dashboards, recognition certificates, or even government/institution incentives (tax breaks, compliance credits).**

4. Student Inequality

- **Stronger students (high marks, polished CVs) get picked first; weaker students risk being left behind.**
- **Fix: Institution ensures *fair distribution* by matching students to opportunities, not leaving it entirely to companies.**

5. Data Security & Scams

- **Students applying on scattered sites risk data misuse.**
- **Fix: Centralize applications in a trusted institutional system. Companies must be verified before posting.**

6. Mismatch of Skills

- **Companies post requirements that don't align with students' actual coursework.**
- **Fix: Institution validates postings to ensure they match academic programs. Prevents irrelevant placements.**

7. Graduation Delays

- **Students placed late or not at all → graduation postponed.**

- Fix: Institution tracks placements in real time and intervenes if a student is at risk of missing deadlines.

8. Company Dropouts

- Companies withdraw programs after posting, leaving students stranded.
- Fix: Institution requires formal agreements before posting, so companies commit to hosting students.

9. Student Dropouts

- Some students accept placements but don't show up or quit midway.
- Fix: Institution monitors attendance and completion via placement tracking. Status updates feed into graduation eligibility.

10. Compliance & Legal Risks

- Labor laws, stipends, and POPIA (data protection) must be respected.
- Fix: Institution acts as compliance gatekeeper, ensuring postings meet legal standards before students are matched.

🎯 **The Truth**

If you don't design for these loopholes, your system will just replicate the same frustrations students face today. The key is two-tier eligibility, institutional verification, company incentives, and compliance tracking. That way, students don't miss cycles, companies see value, and institutions keep graduation rates intact.

👉 Do you want me to prioritize these loopholes into "critical to fix now" vs "nice to fix later" so you know where to focus your design first?